Planet-Constrained Hydrothermal Vent Chemistry Coupled to PHREEQC and Protocell Evolution

LIFEORIG working model specification

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1 Purpose

This document defines a simulation workflow in which planetary parameters determine the physical and chemical boundary conditions for an ocean–hydrothermal vent system. PHREEQC is then used to compute local aqueous speciation, pH, ionic strength, redox state, and mineral saturation. The output of PHREEQC feeds a non-equilibrium organic kinetic network that produces reduced carbon species, activated intermediates, amino-acid precursors, fatty acids, peptides, and compartment-like structures. These products are then used to define a probabilistic protocell nucleation and evolution layer.

The central pipeline is

$$\boxed{\text{Planet model} \rightarrow \text{bulk ocean/vent chemistry} \rightarrow \text{PHREEQC speciation} \rightarrow \text{organic kinetic network} \rightarrow \text{protocell dyn}} \tag{1}$$

The key modeling rule is:

> PHREEQC does not create the organic network. PHREEQC provides the local chemical environment in which the organic network evolves.

PHREEQC is used as the geochemical equilibrium/speciation engine because it can calculate aqueous species distributions, saturation indices, reaction paths, and one-dimensional transport/inverse geochemical calculations [1, 2, 3].

2 Current implementation status

The current repository contains a PHREEQC smoke test in:

`src/ocean_chem/test_ocean_chem_0.py`

The test imports `phreeqpy`, loads:

`external/phreeqc/database/phreeqc.dat`

and runs a simple NaCl solution with selected output for pH, temperature, ionic strength, Na, and Cl. At the time this note was added, `phreeqpy` was available in the local project virtual environment `env/bin/python`. The conda environment used for other work may need a clean PETSc/`petsc4py`/PHREEQC setup before this backend is folded into the ODE solver stack.

3 Model hierarchy

3.1 Layer 1: planet and geophysical environment

For a water world or icy moon, define a planet object

$$\mathcal{P} = \{R, M, g, d_{\text{ocean}}, d_{\text{ice}}, \Phi_{\text{tidal}}, \Phi_{\text{rad}}, X_{\text{rock}}, X_{\text{ice}}, \mathcal{C}_{\text{rock}}\}. \tag{2}$$

The model should provide:

$$\begin{aligned} T_{\text{ocean}} &= T_{\text{ocean}}(\mathcal{P}), && (3)\\ T_{\text{vent}} &= T_{\text{vent}}(\Phi_{\text{tidal}}, \Phi_{\text{rad}}, \mathcal{C}_{\text{rock}}), && (4)\\ P(z) &= P_0 + \rho_{\text{ocean}} g z, && (5)\\ F_{\text{H}_2} &= F_{\text{H}_2}(\text{water–rock interaction}), && (6)\\ F_{\text{ox}} &= F_{\text{ox}}(\text{surface radiolysis, oxidant delivery}). && (7) \end{aligned}$$

The physical domain can initially be one-dimensional:

$$x = 0 \quad \text{vent boundary}, \qquad x = L \quad \text{ocean boundary}. \tag{8}$$

A minimal imposed temperature profile is

$$T(x) = T_{\text{vent}} + (T_{\text{ocean}} - T_{\text{vent}})\,\frac{x}{L}. \tag{9}$$

A more realistic model can solve advection–diffusion heat transport:

$$\rho c_p \left(\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x}\right) = \frac{\partial}{\partial x}\left(k_T \frac{\partial T}{\partial x}\right) + Q_{\text{local}}. \tag{10}$$

3.2 Layer 2: PHREEQC geochemistry

For each grid cell x_j, PHREEQC receives:

$$\text{PHREEQC input}(x_j) = \{T_j, P_j, \text{pH}_{\text{guess}}, \text{pe}_{\text{guess}}, \mathbf{b}_j, \mathcal{M}_j, \mathcal{G}_j\}, \tag{11}$$

where $\mathbf{b}_j$ is the vector of bulk elemental totals, $\mathcal{M}_j$ is the set of allowed mineral phases, and $\mathcal{G}_j$ is the optional gas phase.

PHREEQC returns:

$$\text{geo}_j = \{\text{pH}_j, \text{pe}_j, I_j, a_i(x_j), C_i(x_j), \text{SI}_m(x_j)\}. \tag{12}$$

Here a_i are activities, C_i are aqueous concentrations, and SI_m are mineral saturation indices.

3.3 Layer 3: organic kinetic network

The organic species evolve by reaction–transport:

$$\frac{\partial C_i^{\text{org}}}{\partial t} = -\frac{\partial}{\partial x}(u C_i^{\text{org}}) + \frac{\partial}{\partial x}\left(D_i \frac{\partial C_i^{\text{org}}}{\partial x}\right) + R_i(\mathbf{C}^{\text{org}}, \text{geo}, T, \mathcal{M}). \tag{13}$$

Each reaction rate is evaluated using PHREEQC outputs:

$$r_\alpha = k_\alpha(T) \prod_i a_i^{\nu_{i\alpha}}\, f_\alpha(\text{pH})\, g_\alpha(I)\, h_\alpha(\text{minerals})\, q_\alpha(\text{redox}). \tag{14}$$

The temperature-dependent prefactor may be represented as

$$k_\alpha(T) = k_{\alpha,0} \exp\left(-\frac{E_\alpha}{RT}\right). \tag{15}$$

Mineral catalysis can be represented initially as a multiplicative factor:

$$h_\alpha(\text{minerals}) = 1 + \sum_m \eta_{\alpha m} \, A_m \, \Theta_m, \tag{16}$$

where A_m is the reactive mineral surface area and Θ_m is an availability factor derived from saturation state or explicit mineral abundance.

3.4 Layer 4: protocell nucleation and evolution

The organic network produces coarse pools:

$$\mathbf{O} = \{C_{\text{formate}}, C_{\text{acetate}}, C_{\text{pyruvate}}, C_{\text{thioester}}, C_{\text{acetylP}}, C_{\text{AA}}, C_{\text{peptide}}, C_{\text{FA}}, C_{\text{vesicle}}\}. \tag{17}$$

A local protocell nucleation score is

$$S_{\text{proto}} = w_L \log(1+C_{\text{FA}}) + w_P \log(1+C_{\text{peptide}}) + w_E \log(1+C_{\text{thioester}}+C_{\text{acetylP}}) + w_G \Delta\mu_{\text{redox}} + w_S S_{\text{surface}} - \Pi_{\text{stress}}. \tag{18}$$

The probability of nucleation in cell x_j during Δt is

$$P_{\text{proto}}(x_j, \Delta t) = 1 - \exp\left[-\lambda_0 \Delta t \, \sigma(S_{\text{proto}})\right], \qquad \sigma(S) = \frac{1}{1+e^{-S}}. \tag{19}$$

The stress penalty should include:

$$\Pi_{\text{stress}} = b_T \Pi_T + b_{\text{pH}} \Pi_{\text{pH}} + b_{\text{div}} \Pi_{\text{Mg/Ca}} + b_{\text{dil}} \Pi_{\text{dilution}} + b_{\text{hyd}} \Pi_{\text{hydrolysis}}. \tag{20}$$

4 Initial conditions for water worlds

The following numbers are not final observational constraints. They are scenario templates for simulation design. Europa and Ganymede ocean compositions are uncertain; published models commonly explore chloride-rich, sulfate-rich, carbonate-bearing, and water–rock alteration scenarios [7, 8, 9, 10]. The values below should therefore be treated as tunable hypotheses.

4.1 Scenario A: Europa-like oxidant-influenced chloride/sulfate ocean

Quantity	Symbol	Starting value / range
Ocean temperature	T_{ocean}	260 K to 290 K
Vent temperature	T_{vent}	320 K to 500 K
Seafloor pressure	P_{sf}	50 MPa to 150 MPa
pH initial guess	pH	5.5–9.5
Na	Na	0.05–0.70 mol/kgw
Cl	Cl	0.05–0.70 mol/kgw
Mg	Mg	10^{-3}–0.1 mol/kgw

Ca	Ca	10^{-4}–0.05 mol/kgw
Total carbonate	C(4)	10^{-4}–10^{-1} mol/kgw
Sulfate	S(6)	10^{-5}–10^{-1} mol/kgw
Sulfide	S(-2)	10^{-8}–10^{-3} mol/kgw
Fe	Fe(2)	10^{-9}–10^{-4} mol/kgw
Ni	Ni(2)	10^{-10}–10^{-6} mol/kgw
Ammonia/ammonium	N(-3)	10^{-8}–10^{-4} mol/kgw
Phosphate	P	10^{-9}–10^{-5} mol/kgw
Silica	Si	10^{-5}–10^{-2} mol/kgw
Hydrogen	H2(aq)	10^{-8}–10^{-2} mol/kgw
Methane	CH4(aq)	10^{-9}–10^{-3} mol/kgw

4.2 Scenario B: Ganymede-like more reduced ocean

Quantity	Symbol	Starting value / range
Ocean temperature	T_{ocean}	250 K to 290 K
Vent temperature	T_{vent}	310 K to 450 K
Seafloor pressure	P_{sf}	80 MPa to 250 MPa
pH initial guess	pH	7.0–11.5
Na	Na	0.01–0.50 mol/kgw
Cl	Cl	0.01–0.50 mol/kgw
Mg	Mg	10^{-4}–0.05 mol/kgw
Ca	Ca	10^{-5}–0.02 mol/kgw
Total carbonate	C(4)	10^{-5}–10^{-1} mol/kgw
Sulfate	S(6)	10^{-8}–10^{-3} mol/kgw
Sulfide	S(-2)	10^{-7}–10^{-2} mol/kgw
Fe	Fe(2)	10^{-9}–10^{-4} mol/kgw
Ni	Ni(2)	10^{-10}–10^{-6} mol/kgw
Ammonia/ammonium	N(-3)	10^{-8}–10^{-4} mol/kgw
Phosphate	P	10^{-10}–10^{-5} mol/kgw
Silica	Si	10^{-5}–10^{-2} mol/kgw
Hydrogen	H2(aq)	10^{-7}–10^{-2} mol/kgw
Methane	CH4(aq)	10^{-8}–10^{-3} mol/kgw

4.3 Scenario C: generic Enceladus-like alkaline hydrothermal ocean

This case is useful as a positive-control scenario for alkaline hydrothermal chemistry.

Quantity	Symbol	Starting value / range
Ocean temperature	T_{ocean}	270 K to 300 K
Vent temperature	T_{vent}	330 K to 500 K
Seafloor pressure	P_{sf}	5 MPa to 50 MPa
pH initial guess	pH	8.5–12.0
Na	Na	0.05–0.80 mol/kgw
Cl	Cl	0.05–0.80 mol/kgw
Carbonate	C(4)	10^{-4}–10^{-1} mol/kgw

Sulfide	S(-2)	10^{-7}–10^{-2} mol/kgw
Hydrogen	H2(aq)	10^{-6}–10^{-1} mol/kgw
Methane	CH4(aq)	10^{-8}–10^{-2} mol/kgw
Silica	Si	10^{-5}–10^{-2} mol/kgw

5 PHREEQC input templates

5.1 Ocean boundary template

```
TITLE Europa-like ocean boundary
SOLUTION 1 Ocean
    temp      {T_ocean_C}
    pressure  {P_ocean_atm}
    units     mol/kgw
    pH        {pH_guess} charge
    pe        {pe_guess}
    Na        {Na}
    Cl        {Cl}
    Mg        {Mg}
    Ca        {Ca}
    C(4)      {C_total}
    S(6)      {Sulfate}
    S(-2)     {Sulfide}
    Fe(2)     {Fe2}
    Ni        {Ni2}
    N(-3)     {NH3_total}
    P         {P_total}
    Si        {Si_total}
    H(0)      {H2_total}
END
```

5.2 Vent boundary template

```
TITLE Reduced hydrothermal vent boundary
SOLUTION 2 Vent
    temp      {T_vent_C}
    pressure  {P_vent_atm}
    units     mol/kgw
    pH        {pH_vent_guess} charge
    pe        {pe_vent_guess}
    Na        {Na_vent}
    Cl        {Cl_vent}
    Mg        {Mg_vent}
    Ca        {Ca_vent}
    C(4)      {C_vent}
    S(-2)     {Sulfide_vent}
    Fe(2)     {Fe2_vent}
    Ni        {Ni2_vent}
```

```
    N(-3)     {NH3_vent}
    P         {P_vent}
    Si        {Si_vent}
    H(0)      {H2_vent}
EQUILIBRIUM_PHASES
    Mackinawite  0.0  0.0
    Pyrite       0.0  0.0
    Magnetite    0.0  0.0
    Brucite      0.0  0.0
    Calcite      0.0  0.0
END
```

6 PHREEQC outputs required by the kinetic network

At each grid cell, extract:

Output	Symbol	Use in kinetic network
pH	pH	Acid/base control, vesicle stability
pe/Eh	pe or E_h	Redox availability
Ionic strength	I	Activity corrections, vesicle stress
CO_2(aq)	a_{CO_2}	Carbon fixation
HCO_3^-	$a_{HCO_3^-}$	Carbonate buffer, carbon source
CO_3^{2-}	$a_{CO_3^{2-}}$	Carbonate precipitation
H_2(aq)	a_{H_2}	Electron donor
CO(aq)	a_{CO}	Carbonyl chemistry
H_2S, HS^-	$a_{sulfide}$	FeS formation, thioester chemistry
Fe^{2+}, Ni^{2+}	a_{Fe}, a_{Ni}	Catalytic mineral availability
NH_3, NH_4^+	a_N	Amino acid precursors
Phosphate species	a_{Pi}	Acetyl phosphate, activation chemistry
Mg^{2+}, Ca^{2+}	a_{Mg}, a_{Ca}	Fatty acid precipitation/stability
Mineral SI	SI_m	Surface availability and precipitation sinks

7 Reaction network

The following reaction network is deliberately written as an implementable starting model. Some reactions are elementary-looking; others are lumped effective reactions representing families of pathways. Each reaction should carry a confidence flag and a literature source in the code.

7.1 Module A: aqueous acid–base and geochemical equilibria

These reactions should mostly be handled by PHREEQC, not by the custom kinetic layer.

ID	Reaction	Role
G1	$CO_2(aq) + H_2O \rightleftharpoons HCO_3^- + H^+$	carbonate speciation

G2	$HCO_3^- \rightleftharpoons CO_3^{2-} + H^+$	carbonate speciation
G3	$H_2S \rightleftharpoons HS^- + H^+$	sulfide speciation
G4	$HS^- \rightleftharpoons S^{2-} + H^+$	sulfide speciation
G5	$NH_4^+ \rightleftharpoons NH_3 + H^+$	nitrogen speciation
G6	$H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+$	phosphate speciation
G7	$H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+$	phosphate speciation
G8	$HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+$	phosphate speciation
G9	$Fe^{2+} + HS^- \longrightarrow FeS(s) + H^+$	FeS precipitation
G10	$Ni^{2+} + HS^- \longrightarrow NiS(s) + H^+$	NiS precipitation
G11	$Ca^{2+} + CO_3^{2-} \longrightarrow CaCO_3(s)$	carbonate sink
G12	$Mg^{2+} + {}_2OH^- \longrightarrow Mg(OH)_2(s)$	brucite sink
G13	${}_3Fe^{2+} + {}_4H_2O \longrightarrow Fe_3O_4(s) + {}_8H^+ + {}_2e^-$	magnetite/redox
G14	$FeS(s) + S^0 \longrightarrow FeS_2(s)$	pyrite formation
G15	$SiO_2(aq) \longrightarrow SiO_2(s)$	silica precipitation

7.2 Module B: C1 carbon fixation and redox entry

ID	Reaction	Catalyst / control
R1	$CO_2 + H_2 \longrightarrow HCOO^- + H^+$	FeS/NiS/magnetite
R2	$CO_2 + H_2 \longrightarrow CO + H_2O$	Fe/Ni minerals
R3	$CO + H_2O \longrightarrow CO_2 + H_2$	water–gas shift
R4	$CO_2 + {}_4H_2 \longrightarrow CH_4 + {}_2H_2O$	methanation sink
R5	$HCOO^- + H^+ \rightleftharpoons HCOOH$	PHREEQC/pH
R6	$CO_2 + HS^- + H^+ \longrightarrow COS + H_2O$	sulfide activation
R7	$COS + H_2O \longrightarrow CO_2 + H_2S$	hydrolysis sink
R8	$COS + NH_3 \longrightarrow NH_2CO{-}SH$	carbamoyl/thiocarbamate pool
R9	$CO + H_2S \longrightarrow COS + H_2$	sulfur-carbonyl pool
R10	$HCOO^- + H_2 \longrightarrow HCHO + OH^-$	reduced C1 pool

7.3 Module C: thioester and acetyl-phosphate activation

ID	Reaction	Catalyst / control
R11	$CO + CH_3SH \longrightarrow CH_3COSR$	FeS/NiS surface
R12	$CO_2 + CH_3SH + {}_2H_2 \longrightarrow CH_3COSR + {}_2H_2O$	Fe/Ni sulfides
R13	$CH_3COSR + H_2O \longrightarrow CH_3COO^- + RSH + H^+$	hydrolysis sink
R14	$CH_3COSR + Pi \longrightarrow CH_3COOPO_3^{2-} + RSH$	acetyl phosphate

R15	$CH_3COOPO_3^{2-} + H_2O \longrightarrow CH_3COO^- + Pi + H^+$	hydrolysis sink
R16	$CH_3COOPO_3^{2-} + NH_3 \longrightarrow$ acetamide + Pi	amide side branch
R17	$CH_3COSR + NH_3 \longrightarrow$ thioacetamide + RSH	nitrogen side branch
R18	$CH_3COSR + HCOO^- \longrightarrow$ acetoacetyl_SR	C–C extension
R19	$CH_3COSR + CO_2 + H_2 \longrightarrow$ pyruvate + RSH	C3 precursor
R20	$CH_3COO^- + H^+ \rightleftharpoons CH_3COOH$	PHREEQC/pH

7.4 Module D: C2–C5 central organic precursor network

ID	Reaction	Catalyst / control
R21	HCHO + HCHO $\longrightarrow$ glycolaldehyde	mineral/base catalysis
R22	glycolaldehyde + HCHO $\longrightarrow$ glyceraldehyde	sugar precursor
R23	glycolaldehyde + HCN $\longrightarrow$ glycolonitrile	nitrile branch
R24	HCHO + HCN $\longrightarrow$ glycolonitrile	nitrile branch
R25	glycolonitrile + $_2H_2O \longrightarrow$ glycolate + NH_3	hydrolysis
R26	acetate + $CO_2 + H_2 \longrightarrow$ pyruvate + H_2O	Fe/Ni minerals
R27	pyruvate + $CO_2 + H_2 \longrightarrow$ oxaloacetate	C4 precursor
R28	pyruvate + $H_2 \longrightarrow$ lactate	reduction sink
R29	pyruvate $\longrightarrow$ acetate + CO_2	decarboxylation sink
R30	glyoxylate + acetate $\longrightarrow$ malate	C4 pool abstraction
R31	malate $\longrightarrow$ fumarate + H_2O	dehydration
R32	fumarate + $H_2 \longrightarrow$ succinate	reduction
R33	succinate + $CO_2 + H_2 \longrightarrow$ alpha−ketoglutarate	C5 precursor
R34	alpha−ketoglutarate $\longrightarrow$ succinate + CO_2	degradation sink
R35	oxaloacetate $\longrightarrow$ pyruvate + CO_2	degradation sink

7.5 Module E: amino-acid precursor network

ID	Reaction	Catalyst / control
R36	glyoxylate + $NH_3 + H_2 \longrightarrow$ glycine + H_2O	reductive amination
R37	pyruvate + $NH_3 + H_2 \longrightarrow$ alanine + H_2O	reductive amination
R38	oxaloacetate + $NH_3 + H_2 \longrightarrow$ aspartate + H_2O	reductive amination
R39	alpha−ketoglutarate + $NH_3 + H_2 \longrightarrow$ glutamate + H_2O	reductive amination
R40	HCHO + NH_3 + HCN $\longrightarrow$ aminoacetonitrile + H_2O	Strecker-like
R41	aminoacetonitrile + $_2H_2O \longrightarrow$ glycine + NH_3	hydrolysis
R42	acetaldehyde + NH_3 + HCN $\longrightarrow$ alanine_nitrile + H_2O	Strecker-like
R43	alanine_nitrile + $_2H_2O \longrightarrow$ alanine + NH_3	hydrolysis

R44	amino_acid + $H_2O \longrightarrow$ degradation_products	hydrolysis/degradation
R45	amino_acid + mineral $\longrightarrow$ amino_acid_surface	retention

7.6 Module F: peptide and activated condensation network

ID	Reaction	Catalyst / control
R46	AA + acetylP $\longrightarrow$ AA^* + acetate	activation
R47	AA^* + AA $\longrightarrow$ $peptide_2$ + Pi	condensation
R48	$peptide_n + AA^* \longrightarrow peptide_{n+1}$ + Pi	chain growth
R49	$peptide_n + H_2O \longrightarrow peptide_{n-1}$ + AA	hydrolysis
R50	$peptide_n$ + mineral $\longrightarrow$ $peptide_n$_surface	retention/catalysis
R51	peptide_surface + FA $\longrightarrow$ peptide_FA_complex	membrane stabilization
R52	$AA^* + H_2O \longrightarrow$ AA + Pi	activation loss
R53	$peptide_n \longrightarrow$ degraded_organic_pool	thermal degradation

7.7 Module G: fatty-acid, amphiphile, and vesicle network

ID	Reaction	Catalyst / control
R54	$CO + H_2 \longrightarrow CH_2^* + H_2O$	Fischer–Tropsch-like
R55	CH_2^* + $acyl\_C_n \longrightarrow acyl\_C_{n+1}$	chain growth
R56	acetyl−SR + $CH_2^* \longrightarrow acyl\_C_3$−SR	chain initiation
R57	$acyl\_C_n$−SR + $H_2O \longrightarrow FA\_C_n$ + RSH	fatty acid release
R58	$FA\_C_n + Mg^{2+} \longrightarrow Mg(FA)_2(s)$	precipitation sink
R59	$FA\_C_n + Ca^{2+} \longrightarrow Ca(FA)_2(s)$	precipitation sink
R60	$FA\_C_n + H^+ \rightleftharpoons HFA\_C_n$	pH-dependent protonation
R61	$FA\_C_n \longrightarrow$ micelle	self-assembly
R62	micelle + $FA\_C_n \longrightarrow$ vesicle	growth
R63	vesicle + $FA\_C_n \longrightarrow$ larger_vesicle	membrane growth
R64	vesicle + peptide $\longrightarrow$ stabilized_vesicle	stabilization
R65	vesicle $\longrightarrow$ micelle + FA_loss	disruption
R66	stabilized_vesicle + organics $\longrightarrow$ protocell_candidate	protocell nucleation

8 Rate-law templates

For each reaction α, store:

```
{
  "id": "R1",
```

```
  "reactants": {"CO2": 1, "H2": 1},
  "products": {"formate": 1, "H+": 1},
  "phase": "surface_aqueous",
  "catalysts": ["FeS", "NiS", "magnetite"],
  "k0": null,
  "Ea_kJ_mol": null,
  "pH_window": [8.0, 11.5],
  "T_window_K": [273, 500],
  "depends_on_phreeqc": ["CO2", "H2", "pH", "pe", "FeS_SI", "NiS_SI"],
  "confidence": "medium",
  "reference": ["MartinRussell2006", "Sojo2016"]
}
```

Use activities where possible:

$$r_{\mathrm{CO_2 \to formate}} = k_1(T) a_{\mathrm{CO_2}} a_{\mathrm{H_2}} h_{\mathrm{FeS/NiS}} f_{\mathrm{pH}} f_{\mathrm{redox}}. \tag{21}$$

For fatty-acid vesicle formation:

$$r_{\mathrm{ves}} = k_{\mathrm{ves}} C_{\mathrm{FA}} f_{\mathrm{pH}}(\mathrm{pH}) f_I(I) f_{\mathrm{div}}(a_{\mathrm{Mg^{2+}}}, a_{\mathrm{Ca^{2+}}}), \tag{22}$$

with f_{div} decreasing when divalent cations precipitate fatty acids.

9 Numerical algorithm

1. Build planet object and compute $T(x)$, $P(x)$, flow velocity $u(x)$, and energy/redox fluxes.
2. Initialize bulk element totals for ocean and vent endmembers.
3. For each grid cell, mix ocean and vent endmembers:

$$b_i(x) = \lambda(x) b_i^{\mathrm{vent}} + [1 - \lambda(x)] b_i^{\mathrm{ocean}}. \tag{23}$$

4. Run PHREEQC at each grid cell using $T(x)$, $P(x)$, $b_i(x)$, gas constraints, and mineral phases.
5. Extract pH, pe, ionic strength, activities, and mineral saturation indices.
6. Evaluate kinetic rates r_α for the organic network using PHREEQC output.
7. Update organic concentrations by reaction–transport.
8. Feed selected organic totals back into PHREEQC only if they significantly affect charge balance, complexation, or precipitation.
9. Compute local protocell nucleation probability.
10. Create, evolve, divide, or destroy protocell agents.

10 Python object interfaces

10.1 Planet state

```
@dataclass
class PlanetState:
    name: str
    radius_m: float
    mass_kg: float
    gravity_m_s2: float
    ocean_depth_m: float
    ice_shell_thickness_m: float
    tidal_heat_flux_W_m2: float
    radiogenic_heat_flux_W_m2: float
    oxidant_flux_mol_m2_s: float
    rock_model: str
```

10.2 Geochemical cell input

```
@dataclass
class GeoCellInput:
    T_K: float
    P_Pa: float
    totals_mol_kgw: dict
    pH_guess: float
    pe_guess: float
    minerals: list
    gases: dict
```

10.3 PHREEQC output state

```
@dataclass
class GeoState:
    pH: float
    pe: float
    ionic_strength: float
    activities: dict
    concentrations: dict
    saturation_indices: dict
    mineral_amounts: dict
```

10.4 Organic network state

```
@dataclass
class OrganicState:
    formate: float
    formaldehyde: float
    CO: float
    acetate: float
```

```
    pyruvate: float
    oxaloacetate: float
    acetyl_thioester: float
    acetyl_phosphate: float
    amino_acid_pool: float
    activated_amino_acid_pool: float
    peptide_pool: float
    fatty_acid_pool: float
    micelle_pool: float
    vesicle_pool: float
```

10.5 Protocell agent

```
@dataclass
class Protocell:
    position: float
    lipid_inventory: float
    peptide_inventory: float
    energy_inventory: float
    permeability: float
    stability: float
    catalyst_genotype: dict
    size: float
    age: float
```

11 References

References

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